

ELECTRONICS STARTER KIT MANUAL

LEARN • BUILD • CREATE

EXPLORE THE WORLD OF
ELECTRONICS AND ROBOTICS
THROUGH HANDS-ON LEARNING AND
CREATIVE EXPERIMENTATION.



Starter Kit

What is a Robotics Starter Kit ?

A robotics starter kit is a beginner-friendly set of tools and components that helps students learn how to build simple robots and electronic projects.

Inside the kit, you will usually find:

- **Electronic parts** (LEDs, resistors, sensors)
- **Movement parts** (motors, wheels)
- **Connecting items** (wires, breadboard)
- **Power source** (battery)

Why is a Robotics Starter Kit Used?

1. To Learn Basic Concepts: Students understand important ideas like:

- How electricity flows
- How circuits are made
- How machines move

2. Hands-On Learning: Instead of only reading, students:

- Build real projects
- Try things on their own
- Learn by doing

3. To Build Simple Robots: Using the kit, students can create:

- LED light circuits
- Light or sound-based projects

4. Improves Problem-Solving Skills: While building, students:

- Make mistakes and fix them
- Think step by step
- Learn how to solve problems

5. Strong Foundation for Future: It prepares students for:

- Advanced robotics
- Engineering and technology fields

Safety Instructions

- Always check connections before powering the circuit
- Do not short-circuit batteries
- Use correct polarity for LEDs and batteries
- Avoid touching exposed wires when powered
- Use tools carefully (wire stripper, screwdriver)
- Do not exceed component voltage ratings

Kit Components Overview

LED (Light Emitting Diode):An LED is a small light that glows when electricity passes through it. It has two legs: a longer positive leg and a shorter negative leg. It is used in indicators, decorations, and simple circuits.

RGB LED :An RGB LED combines Red, Green, and Blue LEDs in one package to produce multiple colors.By varying the intensity of each color, you can generate millions of color combinations.

Buzzer:A buzzer is a component that produces sound when powered. It is commonly used in alarms and alert systems. It works easily with a battery in simple circuits.

Vibrator Motor:A vibrator motor creates vibrations instead of smooth rotation. It is used in mobile phones and small toys. It works when electricity is supplied.

Transistor:A transistor is used to control or switch electrical current. It has three legs and can act like a small switch. It is important in many electronic circuits.

Diode:A diode allows current to flow in only one direction. It helps protect circuits from wrong connections. It has a marked side to show the negative terminal.

Capacitor:A capacitor stores and releases small amounts of electrical energy. It charges and discharges in a circuit. It is used to smooth power supply and signals.

Potentiometer (Variable Resistor):A potentiometer is a resistor that can change its value by turning a knob. It is used to control volume, brightness, or speed. It has three terminals.

Conductive Copper Tape:Copper tape is used as a flat wire in paper circuits. It conducts electricity and sticks to surfaces easily. It is useful for creative electronics projects.

Push Button:A push button is a switch that works only when pressed. It completes the circuit temporarily. It is used in bells, toys, and simple devices.

Slide Switch:A slide switch turns a circuit ON or OFF by sliding it. It stays in position after switching. It is commonly used in small electronic devices.

Toggle Switch:A toggle switch uses a lever to turn power ON or OFF. It stays in its position until changed. It is used in many electrical systems.

Rocker Switch:A rocker switch works by pressing one side to turn ON and the other to turn OFF. It is commonly used in appliances. It is easy to operate.

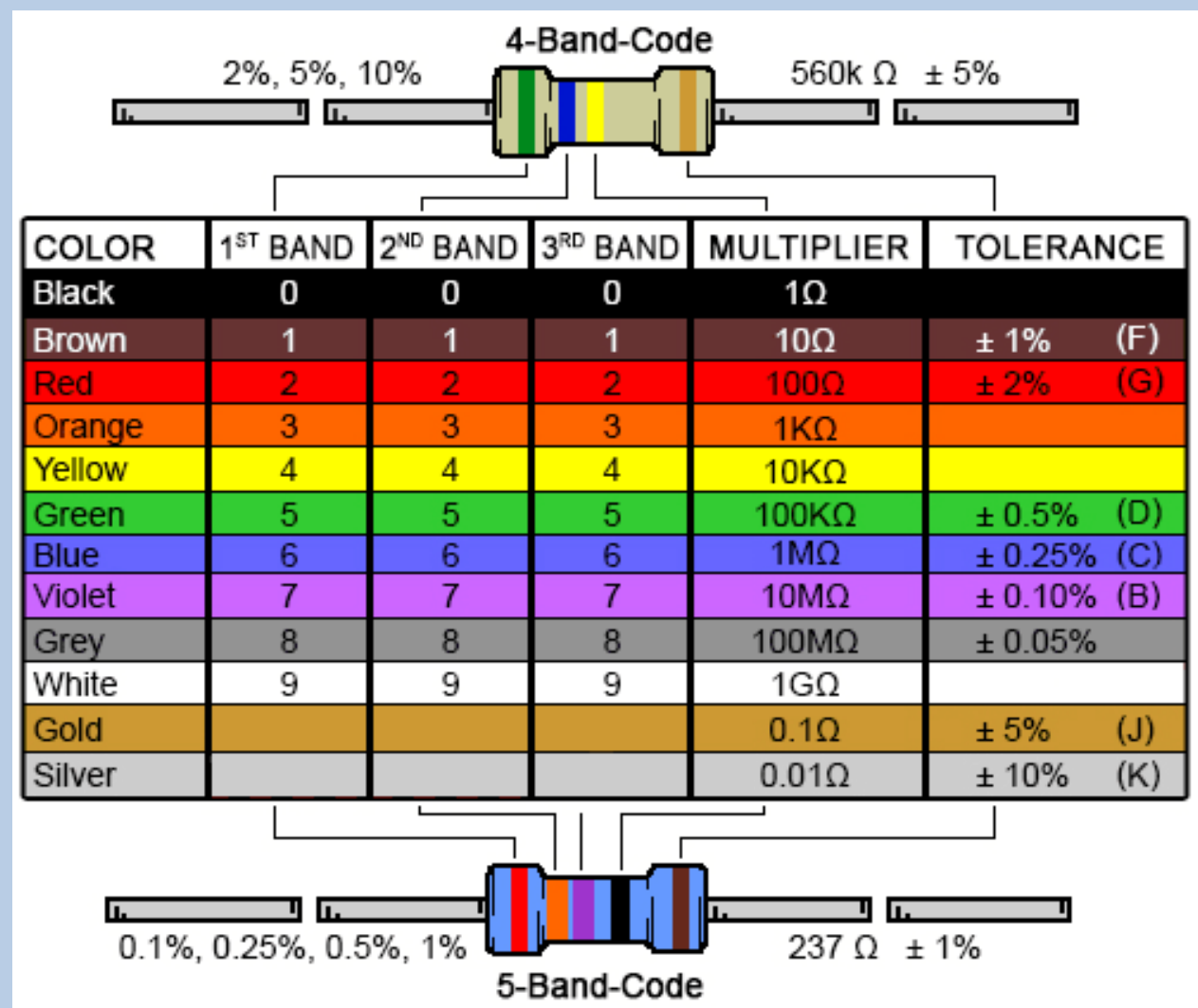
Battery:A battery provides electrical energy to a circuit. It has positive and negative terminals. It is required to power electronic devices.

Battery Clip:A battery clip connects the battery to the circuit safely. It holds the battery firmly in place. It makes wiring easier.

Jumper Wire:Jumper wires are used to connect components in a circuit. They are easy to use with breadboards. They help in quick testing and building.

Coin Battery:A coin battery is a small, flat battery. It provides low voltage, usually 3V. It is used in watches and paper circuits.

Resistor:A resistor controls the amount of current flowing in a circuit. It protects components like LEDs from too much current. Its value is shown using color bands.



Band order for 4 band:

1st Band → 1st digit

2nd Band → 2nd digit

3rd Band → Multiplier

4th Band → Tolerance

Example: Red – Violet – Yellow – Gold

- Red = 2
- Violet = 7
- Yellow = $\times 10,000$
- Gold = $\pm 5\%$

Value = $27 \times 10,000 = 270,000 \Omega = 270 \text{ k}\Omega \pm 5\%$

Band order for 5 band:

1st, 2nd, 3rd = digits

4th = multiplier

5th = tolerance

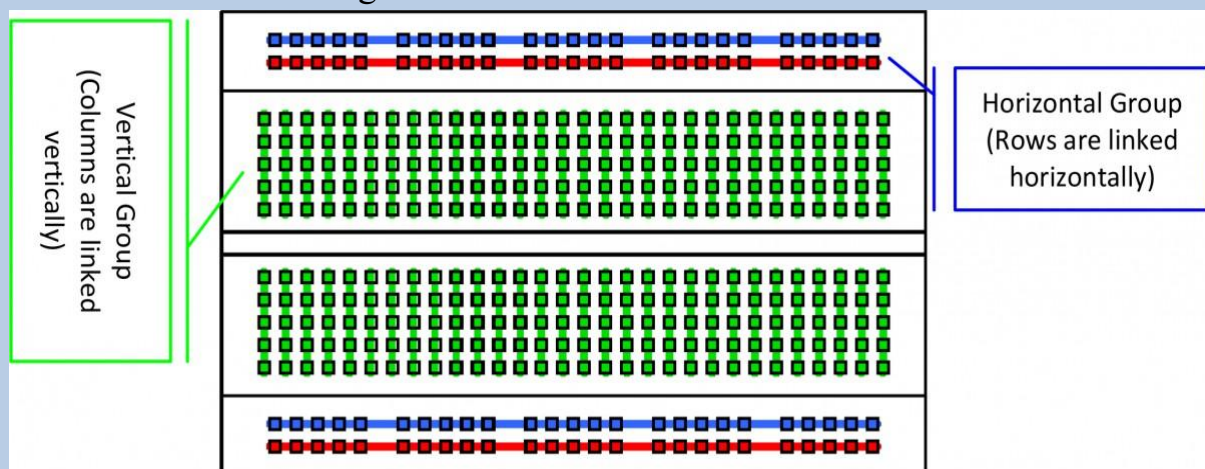
Example: Brown – Black – Red – Orange – Brown

- Brown = 1
- Black = 0
- Red = 2
- Orange = $\times 1,000$
- Brown = $\pm 1\%$

Value = $102 \times 1,000 = 102,000 \Omega = 102 \text{ k}\Omega \pm 1\%$

Breadboard:

A breadboard is used to build circuits without soldering. Components can be inserted and removed easily. It is useful for learning and experimenting. A breadboard is used to build circuits without soldering. Inside it, metal strips connect certain holes together.





Projects

1. Led Glow

Components

- LED
- Resistor
- Battery
- Wires
- Bread Board

Connections

1. Battery

- 9V Battery (+) → Connected to positive rail (red line) on breadboard
- 9V Battery (−) → Connected to negative rail (ground)

2. LED

- Anode (+ longer leg) → Connected to positive supply (through wire)
- Cathode (– shorter leg) → Connected to resistor

3. Resistor

- One end → Connected to LED cathode
- Other end → Connected to ground (– rail)

Current Path

Battery (+) → LED → Resistor → Battery (–)

Working

1. Power Supply

- When the battery is connected, voltage is applied across the circuit.

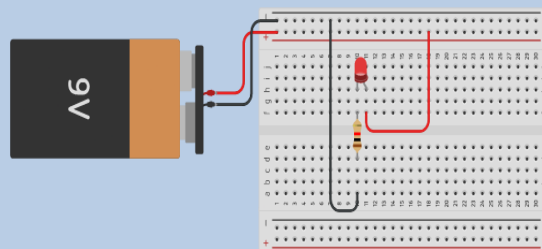
2. LED Operation

- Current flows from positive → LED → resistor → ground
- The LED gets forward biased and starts glowing

3. Role of Resistor

- Limits current flowing through LED
- Prevents LED from burning out

Reference Images



2. Led Glow with Paper Circuit

Components

- LED
- Resistor
- Battery

Connections

1. Battery (Coin Cell)

- Place the coin cell on the copper tape
- Top side (+) → Connected to top copper tape line
- Bottom side (−) → Connected to bottom copper tape line

2. Copper Tape

- Two strips act like wires:
 - Top strip → Positive (+)
 - Bottom strip → Negative (−)

3. LED Connection

- Long leg (Anode +) → Connected to top copper tape
- Short leg (Cathode −) → Connected to bottom copper tape

Complete Path

Battery (+) → Copper Tape → LED → Copper Tape → Battery (−)

Working

Step 1: Power Supply

- The coin cell provides voltage to the copper tape.

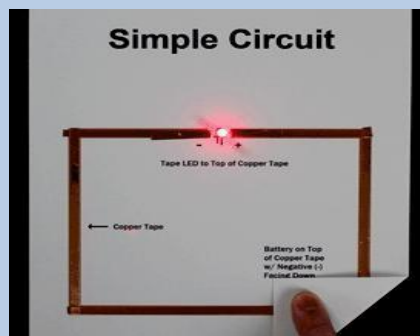
Step 2: Circuit Completion

- When battery touches both copper strips:
 - Circuit becomes closed

Step 3: LED Glows

- Current flows through LED
- LED turns ON

Reference Images



3. Led Glow with Switch

Components

- LED
- Resistor
- Switch
- Battery

Connections

1. Battery

- 9V Battery (+) → Connected to push button input
- 9V Battery (-) → Connected to ground rail

2. Push Button

- One side of push button → Connected to battery (+)
- Other side → Connected to LED (through resistor)

3. LED

- Anode (+ long leg) → Connected to push button output
- Cathode (- short leg) → Connected to resistor

4. Resistor

- One end → Connected to LED cathode
- Other end → Connected to ground (- rail)

Complete Path

Battery (+) → Push Button → LED → Resistor → Battery (-)

Working

Step 1: Button NOT Pressed

- Circuit is open
- No current flows
- LED is OFF

Step 2: Button Pressed

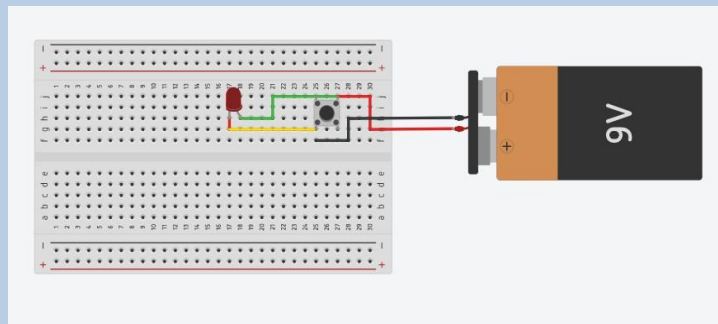
- Push button closes the circuit
- Current starts flowing

Step 3: LED Glows

- Current flows through LED
- LED turns ON only while pressing



Reference Image



3. Alarm with switch

Components

- Buzzer
- Switch
- Battery

Connections

1. Battery

- 9V Battery (+) → Connected to push button input
- 9V Battery (−) → Connected to ground rail

2. Push Button

- One side → Connected to battery (+)
- Other side → Connected to buzzer (+ terminal)

3. Buzzer

- Positive (+) → Connected to push button output
- Negative (-) → Connected to ground (battery -)

Complete Path

Battery (+) → Push Button → Buzzer → Battery (-)

Working

Step 1: Button NOT Pressed

- Circuit is open
- No current flows
- Buzzer is OFF (no sound)

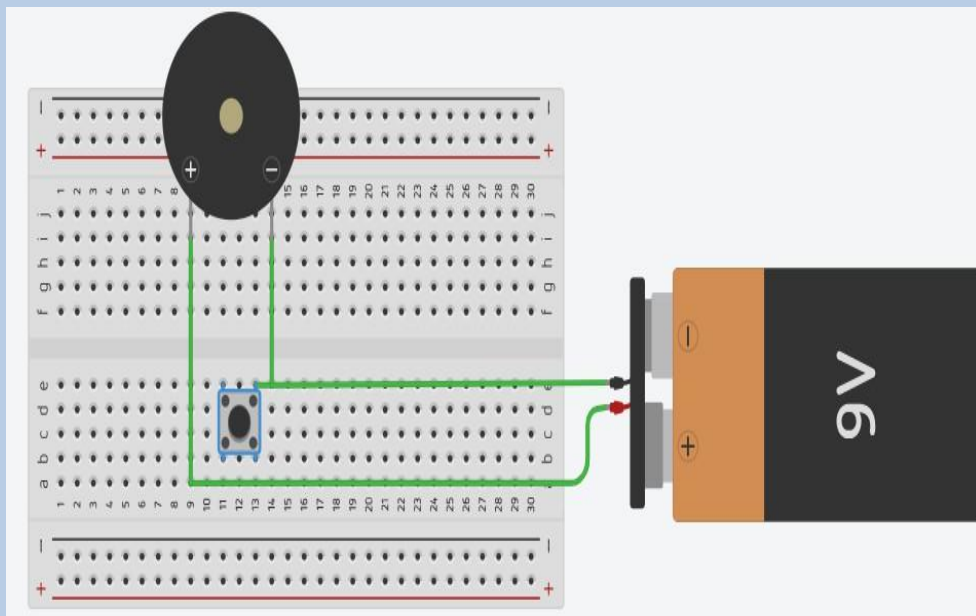
Step 2: Button Pressed

- Circuit becomes closed
- Current flows through buzzer

Step 3: Sound Produced

- Buzzer converts electrical energy into sound
- You hear a beep sound

Reference Image



4. Two way switch connection

Components

- Buzzer
- Switch - 2
- Battery
- Jumper wire

Connections

1. Battery

- 9V Battery (+) → Connected to positive rail (red line)
- 9V Battery (–) → Connected to ground rail

2. Switches (Slide + Toggle)

- Battery (+) → Connected to first switch (slide switch)
- Output of slide switch → Connected to second switch (toggle switch)
- Output of second switch → Goes to LED

(Switches are connected in series)

3. LED

- Anode (+ long leg) → Connected to output of switches
- Cathode (– short leg) → Connected to resistor

4. Resistor

- One end → Connected to LED cathode
- Other end → Connected to ground (– rail)

Complete Path

Battery (+) → Switch 1 → Switch 2 → LED → Resistor → Battery (–)

Working

1. Any Switch OFF

- Circuit is open
- Current cannot flow
- LED remains OFF

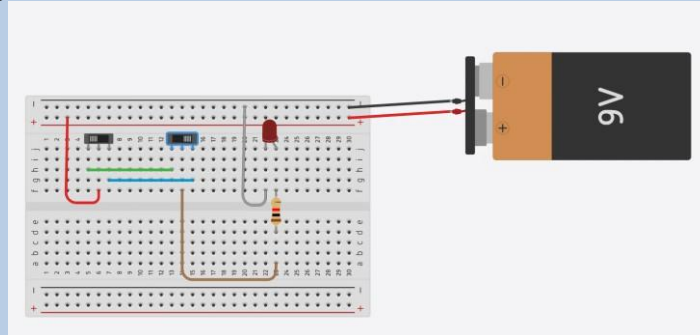
2. Both Switches ON

- Circuit becomes closed
- Current flows through LED

3. LED Turns ON

- LED glows only when both switches are ON

Reference Image



5. Traffic light using LED

Components

- Resister
- LED- 3
- Switch
- Battery
- Jumper wire

Connections

1. Battery

- 9V Battery (+) → Connected to positive rail
- 9V Battery (-) → Connected to ground rail

2. LEDs (3 LEDs)

Each LED is connected in a separate branch:

- Anode (+) → Connected to its respective push button
- Cathode (-) → Connected to resistor → ground

3. Push Buttons (3 switches)

- One side of each button → Connected to positive rail (+)
- Other side → Connected to respective LED

4. Resistors

- Each LED has its own resistor
- Resistor connects LED cathode to ground

Complete Paths (3 branches)

Battery (+) → Button → LED → Resistor → Battery (–)
(Repeated for each LED)

Working

1. No Button Pressed

- All circuits are open
- All LEDs are OFF

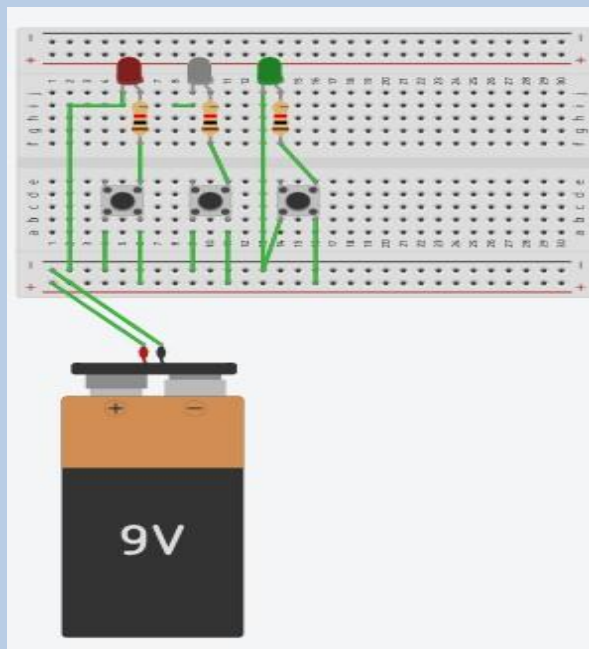
2. Press Any One Button

- That branch becomes closed
- Current flows through that LED
- Only that LED glows

3. Press Multiple Buttons

- Multiple branches close
- Multiple LEDs glow simultaneously

Reference Image



6. Series Lights using paper circuit Components

- LED
- Switch
- Resister
- Copper Tape

Connections

1. Battery (Coin Cell)

- Top side (+) → Connected to top copper tape line
- Bottom side (−) → Connected to bottom copper tape line

2. Copper Tape

- Forms a closed rectangular loop
- Acts like wires:
 - Top path → Positive
 - Bottom path → Negative

3. LEDs (2 LEDs in Series)

- First LED:
 - Anode (+) → Connected to positive copper tape
 - Cathode (−) → Connected to next LED
- Second LED:
 - Anode (+) → Connected to first LED cathode
 - Cathode (−) → Connected to negative copper tape

Complete Path

Battery (+) → LED1 → LED2 → Battery (−)

Working

1. Power Supply

- Coin cell provides voltage to the copper tape

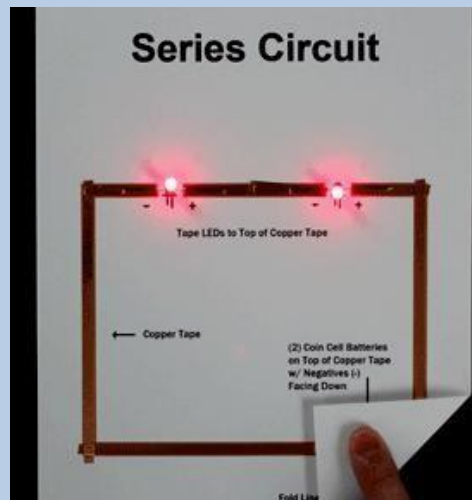
2. Series Connection Behavior

- Current flows through both LEDs one after another
- Same current passes through both LEDs

3. LEDs Glow

- Both LEDs glow together
- If one LED is removed or fails → circuit breaks → both OFF

Reference Image



7. Parallel Lights using paper circuit

Components

- LED
- Switch
- Resister
- Copper Tape

Connections

1. Battery (Coin Cell)

- Top side (+) → Connected to top copper tape line
- Bottom side (-) → Connected to bottom copper tape line

2. Copper Tape Layout

- Two main horizontal lines:
 - Top line → Positive (+)
 - Bottom line → Negative (-)
- Vertical copper paths connect LEDs between these two lines

3. LEDs (Parallel Connection)

Each LED is connected separately:

- Anode (+) → Connected to top copper tape
- Cathode (-) → Connected to bottom copper tape

Each LED has its own path to the battery

Complete Paths

Battery (+) → LED1 → Battery (-)

Battery (+) → LED2 → Battery (-)

(Multiple independent paths)

Working

1. Power Supply

- Coin cell provides voltage across both copper lines

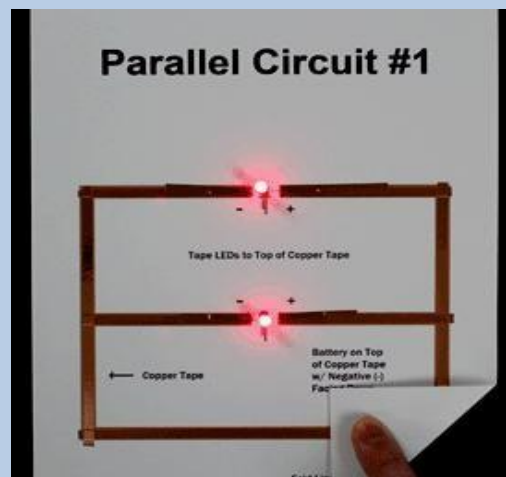
2. Parallel Behavior

- Each LED gets the same voltage
- Current splits into multiple paths

3. LEDs Glow

- All LEDs glow independently
- If one LED fails → others still work

Reference Image



8. Series Lights using LED circuit

Connections (Series Circuit)

1. Battery Connections

- Battery Positive (+) → connected to first LED anode (+)
- Battery Negative (−) → connected to resistor → last LED cathode (−)

2. LED Series Connections

In a series circuit, components are connected end-to-end:

- LED 1 cathode (−) → connected to LED 2 anode (+)
- LED 2 cathode (−) → connected to LED 3 anode (+)
- Continue this chain...

So current flows through all LEDs one after another

3. Resistor Placement

- A single resistor is used in series
- Connected:
 - Either before first LED OR
 - after last LED (as shown in your image)

It limits current for the entire chain

Working of the Circuit

1. Power Flow Starts

- Battery provides voltage
- Current flows from + → through LEDs → resistor → −

2. LEDs Glow Sequentially

- Current passes through LED 1 → LED 2 → LED 3
- Each LED lights up as current flows through it

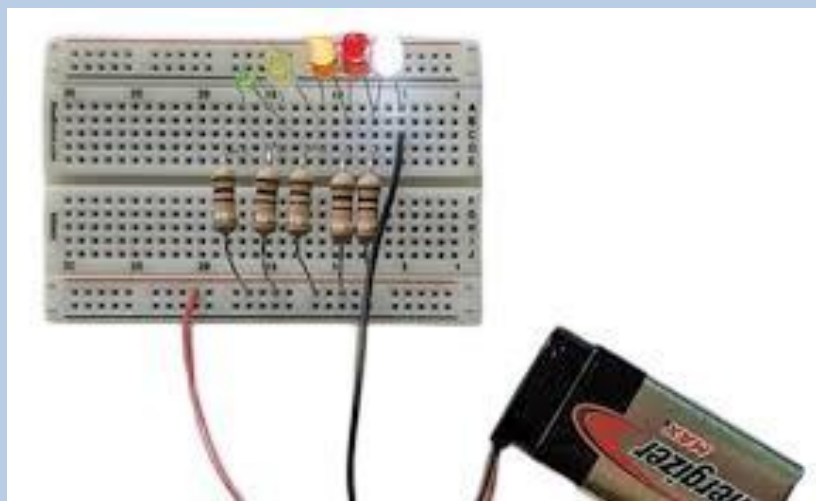
3. Series Behavior

- Same current flows through all components
- Voltage gets divided across LEDs

Example:

- 9V battery with 3 LEDs → each LED gets ~3V (approx)

Reference Image



9. Parallel Lights using LED circuit

Connections

1. Battery Connections

- 9V Battery Positive (+) → Connect to red rail (top power line) on breadboard
- 9V Battery Negative (-) → Connect to blue rail (bottom power line)

2. Resistor Connections (Important for safety)

Each LED must have its own resistor.

- One end of each resistor → connected to positive rail (+)
- Other end of resistor → goes to LED anode (+)

This limits current and protects LEDs.

3. LED Connections

Each LED has two legs:

- Long leg (Anode +)
- Short leg (Cathode −)

Connections:

- Anode (long leg) → connected to resistor output
- Cathode (short leg) → connected to ground rail (−)

4. Parallel Setup

All LEDs are connected in parallel, meaning:

- All anodes connect to + via resistors
- All cathodes connect to − (ground rail)

Working of the Circuit

1. Power Supply Starts

- The 9V battery provides voltage to the breadboard rails
- Current flows from positive → negative

2. Current Through Resistor

- Current first passes through each resistor
- Resistor reduces current to safe levels

3. LED Glows

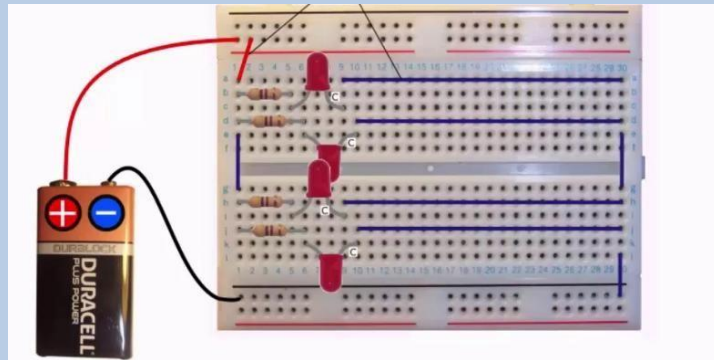
- Current enters LED anode (+)
- Flows through LED → exits via cathode (−)
- LED emits light

4. Parallel Behavior

- Each LED gets same voltage (9V)
- If one LED fails → others still glow



Reference Image



10. LED with Capacitor and transistor

Components

LED

Transistor

Resistor

Battery

Capacitor

Connections

1. Battery Connections

- 9V Battery (+) → Breadboard positive rail
- 9V Battery (-) → Breadboard ground rail

2. LED Connection

- LED Anode (+ longer leg) → Connected to Vcc (+ rail) through a resistor
- LED Cathode (- shorter leg) → Connected to Collector (C) of BC547

3. Transistor

(Flat side facing you → pins are Collector – Base – Emitter)

- Collector (C) → LED negative side
- Emitter (E) → Ground (- rail)
- Base (B) → Connected to resistor + capacitor junction

4. Capacitor

- One terminal → Connected to Base of transistor
- Other terminal → Connected to Ground

5. Resistor (Base Resistor)

- One end → Vcc (+ rail)
- Other end → Base of transistor

6. Extra Resistor (LED protection)

- In series with LED (important to prevent damage)

Working

Step 1: Power ON

- The battery supplies voltage.
- The capacitor starts charging through the resistor.

Step 2: Delay Creation

- Capacitor voltage increases slowly.
- The transistor remains OFF initially.

Step 3: Transistor Turns ON

- When capacitor voltage reaches $\sim 0.7V$:
 - Base gets enough current
 - Transistor switches ON

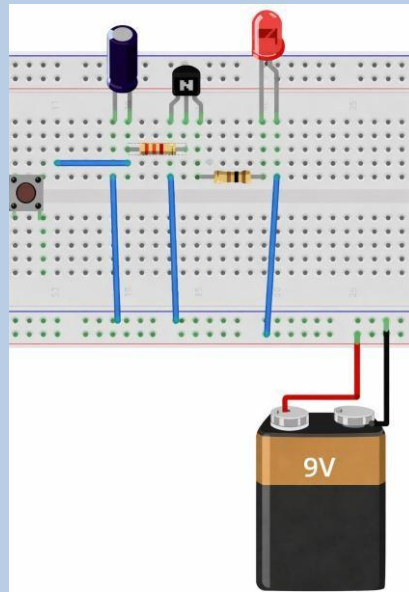
Step 4: LED Glows

- Current flows:
Vcc → LED → Transistor → Ground
- LED turns ON after delay

Step 5: Discharge (if power removed)

- Capacitor discharges → transistor OFF → LED OFF

Reference Image



11. Adjustable LED with Potentiometer

Components

- LED
- Potentiometer
- Battery
- Jumper Wires

Working

The **battery** provides power to the circuit.

- The **LED** lights up when current flows through it.
- The **potentiometer** controls the amount of current going to the LED.

Step-by-step working:

Connection Setup

- One end of the potentiometer → connected to battery (+)
- Other end → connected to ground (-)
- Middle pin (wiper) → connected to the LED (through a resistor if needed)

Adjusting the Knob

- When you rotate the potentiometer knob, it changes resistance.

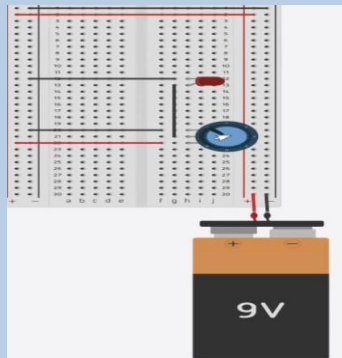
Low Resistance

- More current flows → LED glows bright

High Resistance

- Less current flows → LED becomes dim

Reference Image



12. Working with a Multimeter?

A multimeter is a tool used to measure:

- Voltage (V)
- Current (A)
- Resistance (Ω)
- Continuity (beep test)

Basic Setup

- Black probe → COM (common)
- Red probe → V Ω mA port
- Turn the dial to the required mode

1. Measuring Voltage (Battery)

Steps:

1. Set dial to DC Voltage (V —)
2. Place:
 - Red probe → Positive (+)
 - Black probe → Negative (–)
3. Read value on display

Working:

- Multimeter measures potential difference
- Example: 9V battery → shows around 9V

2. Measuring Resistance (Resistor)

Steps:

1. Set dial to Ω (Ohms)
2. Disconnect resistor from circuit
3. Touch probes on both ends

Working:

- Multimeter sends small current and measures opposition
- Displays resistance value (e.g., 220Ω)

3. Continuity Test

Steps:

1. Set dial to continuity mode
2. Touch probes across wire/component

Working:

- If path is complete $\rightarrow$ beep sound
- If broken $\rightarrow$ no sound

4. Measuring Current (Advanced)

Steps:

1. Move red probe to A or mA port
2. Break circuit and connect multimeter in series
3. Set dial to A

Working:

- Measures flow of current through circuit

Simple Understanding

Voltage = Pressure

Current = Flow

Resistance = Opposition

Safety Tips

- Always start with highest range
- Never measure resistance in a powered circuit
- Check probe connections before measuring current

Package Includes:

1 x Electronics Component Starter Kit

Kit includes:

20 x LED (Light Emitting Diode) – Multicolor (Red, Green, Blue, White, Yellow, RGB), 3mm

1 x Buzzer (6V–12V, 2.8 kHz)

1 x Vibrator Motor (3V DC, 9000 RPM)

5 x BC547 Transistors (NPN)

5 x Diodes

1 x Resistor Kit (30 Values)

5 x Capacitors (Mixed Values)

1 x Potentiometer (10K Ohm)

1 x Copper Tape (10mm Conductive)

3 x Push Buttons (Tactile Switch)

2 x Slide Switch (SPDT)

2 x Toggle Switch

3 x Rocker Switch

1 x 9V Battery

1 x Battery Clip (9V Connector)

60 x Jumper Wires (20 M-M, 20 M-F, 20 F-F)

1 x Breadboard (400 Points)

2 x Coin Cell Batteries (CR2025)

1 x Plastic Storage Box

